Olivia Verge

Mechatronics Engineering Student | Software, Robotics & Embedded Systems

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Education

Memorial University of Newfoundland Sept 2024 – Apr 2029
B.Eng. Mechatronics Engineering (Co-op) *St. John's, NL*

Experience

Nokia Sept 2026 – Dec 2026
Mechanical Hardware Engineer (Co-op) *Ottawa, ON*

- Prototyped Python code to extract materials and item numbers from PDF drawings for a proposed life cycle analysis automation.
- Performed life cycle analysis on 18 product assemblies, reconciling CAD material data against supplier records.

Memorial Medtech May 2026 – present
Software & Electrical Team Member *St. John's, NL*

- Led the front-end design of the team's website for the James Dyson Award application.
- Prototyped the user interface for a portable cardiac triage device in Python (tkinter).
- Researched PPG signal processing and what affects accuracy, such as LED light colour.

Mysa Jan 2026 – May 2026
Hardware Product Engineering Co-op *St. John's, NL*

- Built a full-stack fastener generator that turns plain-English descriptions or photos into STEP, STL, PDF drawing and ZIP files, cutting manual CAD modelling time by about 30%.
- Wrote the FastAPI and CadQuery backend: a natural language parser, geometry for heads, drives and helical threads, and a REST API for chat, image upload and downloads.
- Built a photo pipeline with OpenCV and vision model APIs (Gemini, Groq, OpenAI) that estimates fastener specs, with retries and user confirmation before generating.
- Added Google OAuth, a Next.js/TypeScript shell, Docker deployment and 60+ pytest tests.

Projects

Aftertone: Spotify Listening Companion (in progress) Oct 2026 – present
React, Vite, Python, FastAPI, SQLite

- Building a FastAPI backend with Spotify OAuth and SQLite storage for listening history, with a React dashboard for stats and recommendations.
- Wrote a recommender that ranks tracks from song sequences, artist frequency, time of day and saved places, plus a training script for an optional logistic regression reranker.
- Wrote 20 pytest tests covering the API and recommender.

MUN Engineering Society Website Redesign Jun 2026 – Sept 2026
Next.js, React, Tailwind CSS

- Redesigned the society website (20+ pages) with JSON-driven content so executives can update events and members without touching code.
- Built a contact form API route (Nodemailer) that routes each message to the right society contact.

ClearHead: Concussion Risk Monitor Mar 2026 – Apr 2026
ROS2, Python, JavaScript, WebSockets, Docker

- Built a ROS2 publish/subscribe pipeline that filters normal movement from simulated 6-DOF IMU data, calculates peak linear acceleration and triggers green/yellow/red alerts.
- Built a live JavaScript dashboard with a WebSocket bridge, deployed on GitHub Pages.

Leadership

MUN Engineering Society (Society B) May 2026 – present
President *St. John's, NL*

- Lead a team of 11 (4 VPs, 7 directors); one of four MUN delegates at SDES 2026.
- Previously VP Junior Academic (Jan 2025 – May 2026): organized orientation for 300+ incoming students.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, MATLAB
Frameworks: FastAPI, React, Next.js, Tailwind CSS, ROS2, CadQuery, OpenCV, MediaPipe
Tools & Hardware: Docker, Git/GitHub, pytest, REST APIs, Arduino, Raspberry Pi, IMUs